ordinary reinforcement ($\alpha = 1$) does not lead to an optimal communication system [11].

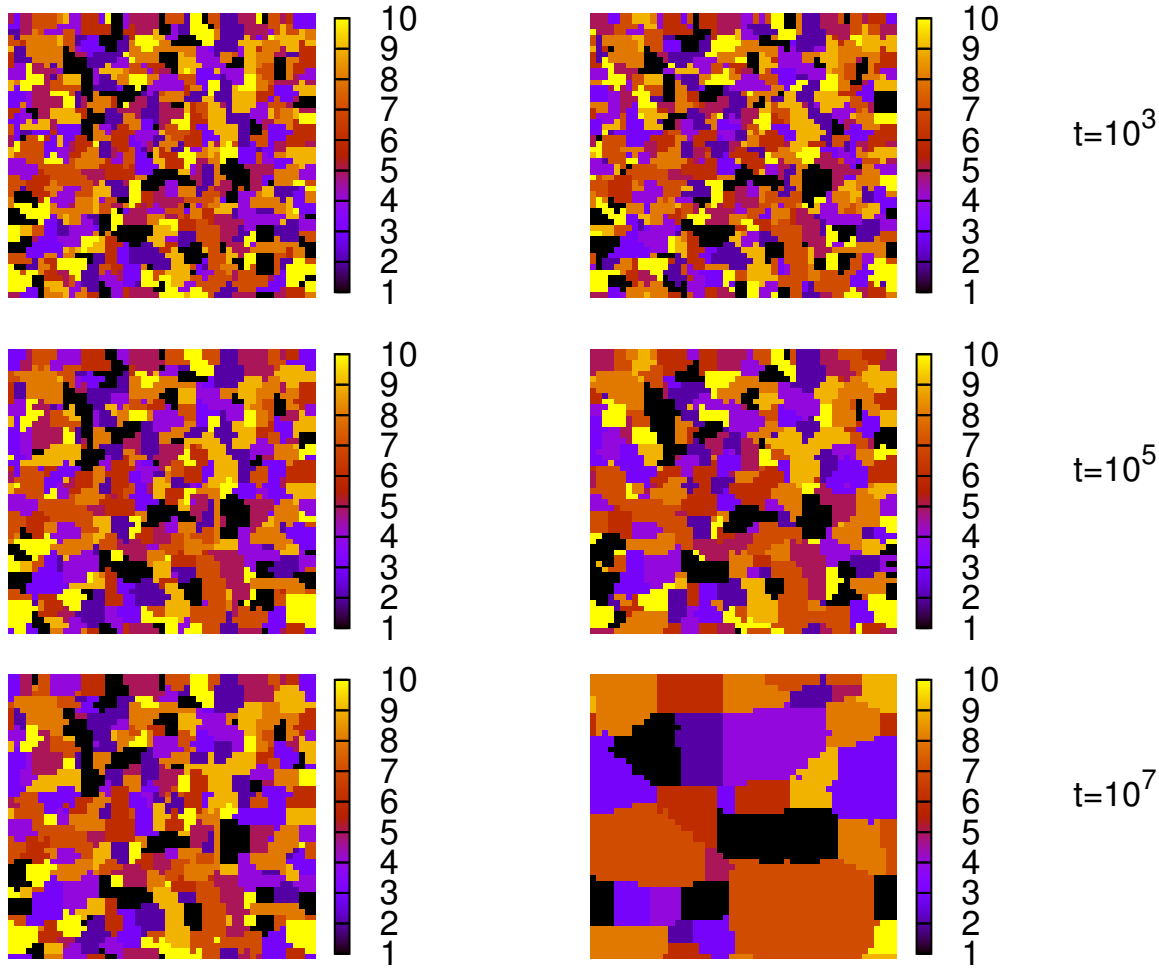

Fig 7. Distribution of the dominant words that agents use to talk about the first object. *Left*: simulations on a square lattice with $N_o = 2$, $N_w = 10$, $N = 50 \cdot 50 = 2.5 \cdot 10^3$, $\alpha = 2$. *Right*: the same simulations but with a population renewal (with probability $p = 10^{-5}$).

Fig 9 provides yet another indication that the dynamics for large and small N_w considerably differ. In this figure, we present a total weight associated with a given word, defined as follows:

$$w_i^{total} = \sum_{A,j} w_{i,j}(A), \quad i = 1, 2, \ldots, N_w \tag{5}$$

where summation is over all agents (A) and objects (j). In Fig 9, what is actually plotted is a normalized total weight given as $w_i^{total} / \sum_k w_k^{total}$. For $N_o = 10$ and $N_w = 50$, we can notice 10 peaks corresponding to the 10 words that are mainly in use. Taking into account a very large success rate (Fig 8), it means that the agents established a single word for each object, which became dominant in their inventories related to this object. As a result only this word is selected by speakers when they decide to talk about the object and just this word leads then to a correct recognition of the object by hearers. The resulting language provides a nearly perfect one-to-one mapping between objects and words. Let us emphasize that such a global language emerges spontaneously, as a result of two-agent interactions only.

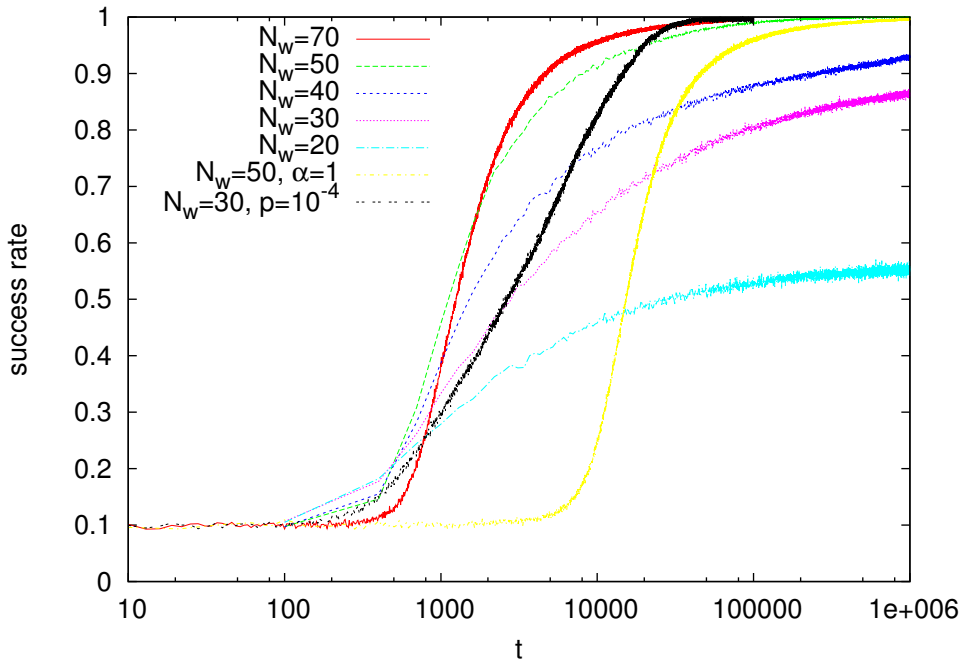

Fig 8. Time dependence of the success rate. Results for the model on the complete graph of size $N = 10^4$ with $N_o = 10$, several values of N_w, and $\alpha = 2$. For $N_w = 50$ and $\alpha = 1$ (yellow line), we can see a much slower convergence to a consensus than for $N_w = 50$ and $\alpha = 2$. The black line shows the success rate for the version with a population renewal (with probability $p = 10^{-4}$).

Much different behavior can be seen for smaller N_w. In Fig 9, some peaks can be also distinguished for $N_w = 30$, but in addition there is an entire spectrum of less important but clearly nonnegligible words, which are being used by agents. The one-to-one correspondence between words and objects is missing in this case and the resulting language has a much smaller success rate (Fig 8). Since agents increase the weights of the communicated word only when the object is recognized correctly, it means that also nondominant words lead sometimes to a correct recognition—otherwise their weights (in relation to dominant words) would diminish to zero. It is an analogue of synonymy, a common feature of natural languages. Synonymy, however, does not reduce the success rate, while homonymy (or polysemy [30]) does. Homonymy appears when a certain word has significant weights associated with more than one object. Communicating such a word may result in an incorrect recognition of the object and the success rate smaller than 1 indicates that homonyms are also present in the emerging language of our model. The structure of our model is quite complex and some intermediate scenarios are also possible. Namely, with respect to some objects, agents may develop a one-to-one relation between objects and words (like for $N_w = 50$), while with respect to some others, a more complex language containing synonyms and homonyms may be used.

Population renewal

As we have already seen, both the single- and multi-object versions of our model on a two-dimensional lattice get trapped in a disordered regime, with only a local consenus reached. In such a regime, the coarsening dynamics, which could lead to formation of larger clusters of agents that reached a consensus, becomes very slow. Some other models with a consensus dynamics, such as the Ising model or the Naming Game, are known to have much faster coarsening dynamics, which could be attributed to the surface tension generated in these dynamics. In this section, we examine our model modified in such a way that the dynamics does not get trapped in a disordered state and induces perhaps some kind of a

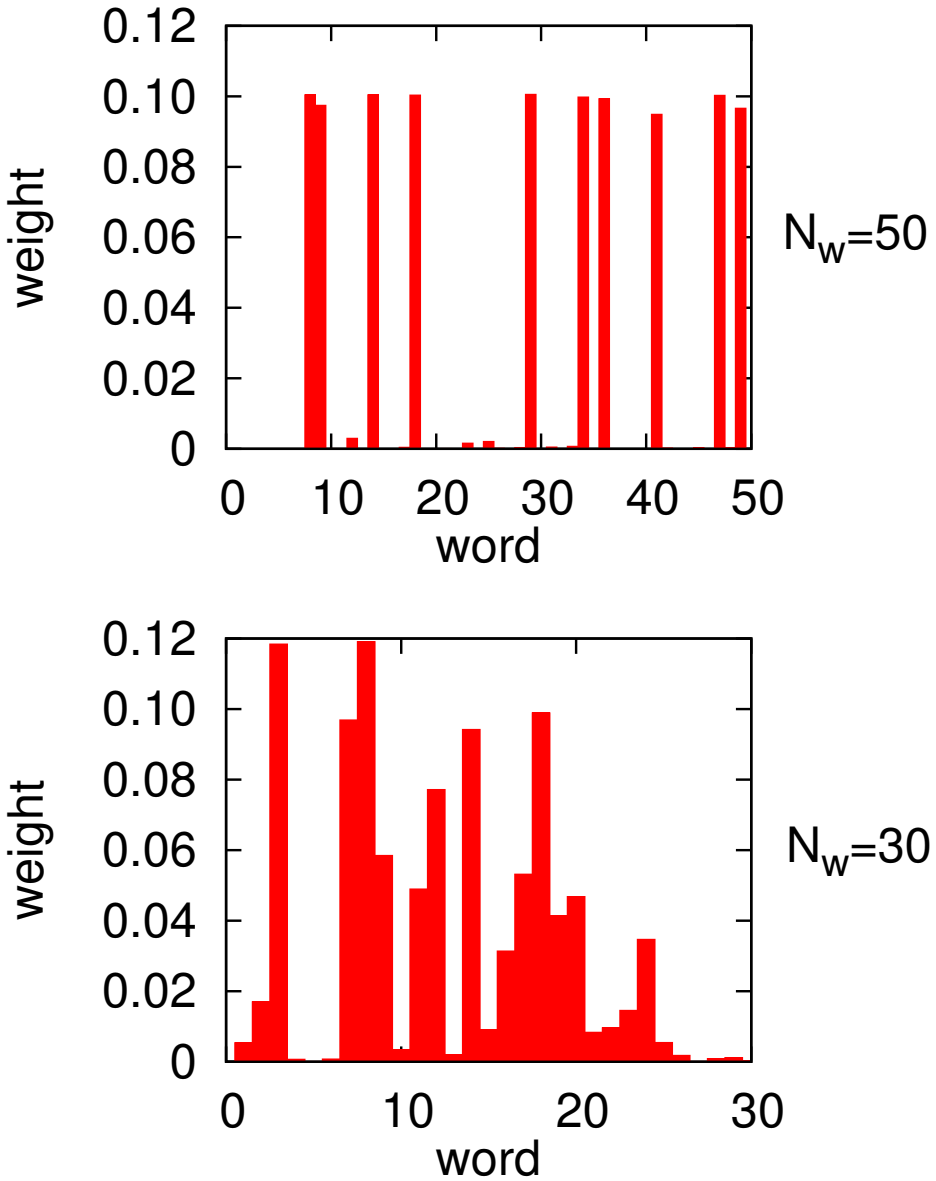

Fig 9. Distribution of the total (normalized) weights associated with particular words. Results for simulations with $N_o = 10$ on the complete graph of size $N = 10^4$ and simulation time $t = 10^6$ ($\alpha = 2$). Simulations for $t = 10^5$ lead to nearly identical distributions.

surface tension. Namely, we introduced a simple mechanism of a population renewal, which means that in each step, the selected agent either (with some probability p) is replaced with a new one (with all weights reset to 1) or else the agent acts as a speaker.

The time evolution of a single-object model with a population renewal on a square lattice is shown in Fig 10. Certainly, the evolution in this case is different than in the absence of population renewal (Fig 3). It seems that there is a tendency to reduce the length of interfaces in this model, just as in some models with a surface tension.

Additional arguments that the dynamics generates some kind of an effective surface tension come from the analysis of time dependence of $1 - m_L$ (Fig 11). Let us notice that significant contributions